Introduction to Quantum Algorithms Based on Group Representations

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Quantum Computing

Black-box Functions

Let R and S be two sets.

A function $f : R \to S$ is called a black-box function if

- we don't know how f is implemented
- the only way we can learn something about f is to query f, i.e., to choose $r \in R$ and to obtain $f(r)$ (possibly multiple times; the queries could be adaptive)

Usually, the task is to determine some property of f.

Black-box Functions on Groups

Let G be a group and S some set.

We say that the black-box function $f : G \to S$ hides a subgroup $H \leq G$ if f satisfies the following two conditions:

- f is constant on the left cosets of H in G, i.e.,

$$f(g) = f(h) \text{ iff } gH = hH$$

- f is different on different cosets, i.e.,

$$f(g) \neq f(h) \text{ iff } gH \neq hH$$

The Stabilizer Problem as a HSP

Important Applications of HSP

omplexity of the Hidden Subgroup Problem

The task is to determine a generating set of the hidden subgroup H by only querying f.

Classically, we need $\Omega(|G|)$ queries to solve the HSP.

In contrast, we can solve the problem with high probability and running time $O(\log^c(|G|))$ on quantum computer for the following families of groups:

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Intro to Quantum Computing

Pontrajagin-duality

Theorem (Pontrajagin-duality)
For every finite abelian group G there is an isomorphism ϕ with $G \cong_\phi \hat{G} := \mathrm{Hom}(G, \mathbb{C}^\times)$.

Let β be the pairing between G and $\hat{G}$ defined by

$$\beta : \begin{cases} G \times \hat{G} & \rightarrow \quad \mathbb{C}^\times \\ (g, \chi) & \mapsto \quad \chi(g) \end{cases}$$

Orthogonal complement of subgroups

We say that $g \in G$ is orthogonal to $\chi \in \hat{G}$, denoted by $g \perp \chi$, if $\beta(g, \chi) = 1$, that is, $g \in \ker(\chi)$.

We identify $\hat{A}$ and A with respect to the isomorphism ϕ and define for a subgroup $H \leq G$ an orthogoanl complement by

$$H^{\perp} := \{y \in G \mid \forall x \in H : y \perp x\}$$

Fourier transform

The Fouriertransform DFT_G over the abelian group G is defined by

$$\mathrm{DFT}_G := \frac{1}{\sqrt{|G|}} \sum_{x,y \in G} \beta(x,y) |y\rangle\langle x|$$

Lemma: For every subgroup $H \leq G$ and $g \in G$ we have

$$\mathrm{DFT}_G \frac{1}{\sqrt{H}} \sum_{x \in H} |g + x\rangle = \frac{1}{\sqrt{|H^\perp|}} \sum_{y \in H^\perp} \beta(g,y) |y\rangle$$